

Data-centric market status, case studies and outlook

Alan Morrison
February 2019



Agenda

Time	Activity
2:00 – 2:10	Introduction
2:10 – 2:20	Larger market context: buzz, trends and dynamics
2:20 – 2:30	Current approaches to the market
2:30 – 2:40	Case studies
2:40 – 3:15	Conclusion and discussion

1

Introduction

Product-centric IT: Got an IT problem? Buy a solution.

Products and services

Data management

Applications

Computer graphics and web design

Development

IT administration

IT certifications

IT security

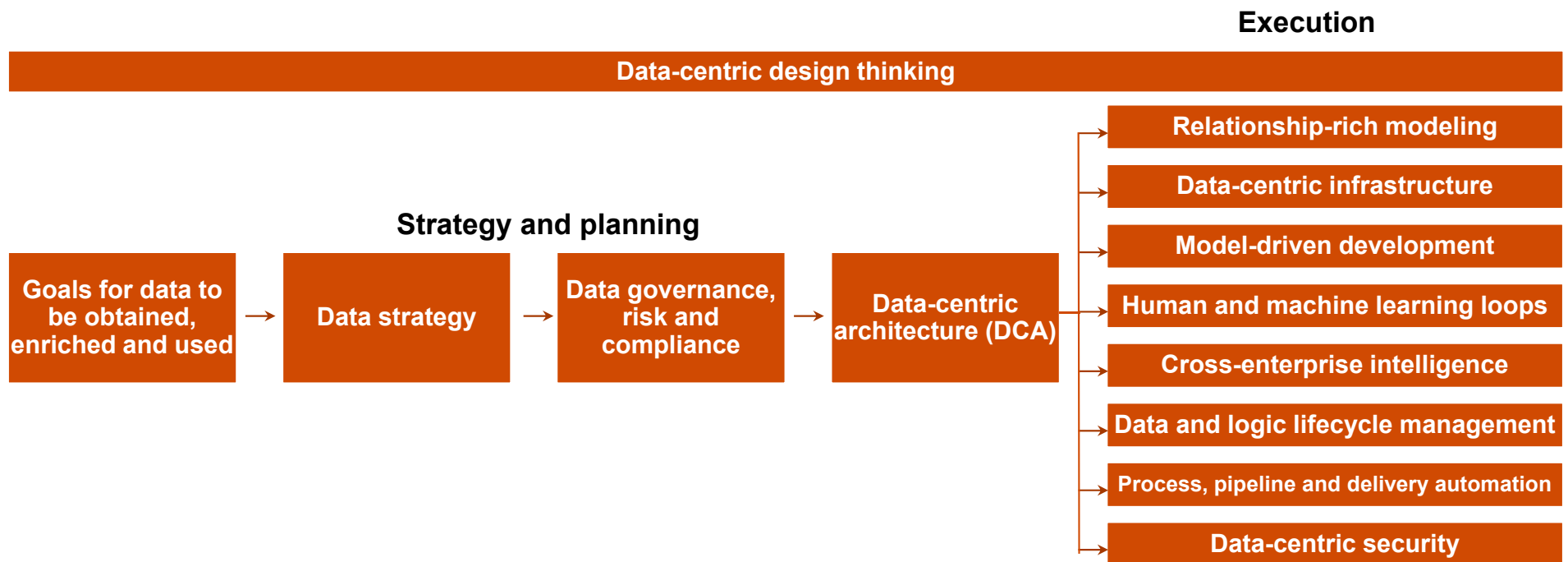
Service providers

Technologies and fields

Categories from wand® precision
classification and search, 2019

- Market oriented, passive, laissez faire
- Architecture and vision free
- “The market will fix it.”
- Big vendor dominated
- Legacy of Nick Carr’s *IT Doesn’t Matter*
- Siloed, isolated efforts
- Startups venture funded in silos, with waves of new silos being generated every year
- Users themselves just *passive and disempowered* buyers or subscribers
- Consumer services mostly self-service, with no one to call if a problem arises

Data-centric IT: Own the problem first. Then build a solution.



- In the data-centric view, every IT category is subordinated to centrally managed, model-driven data via data strategy, GRC and data-centric architecture (DCA)
- Relationship-rich modeling leads development for reasons of efficiency and effectiveness
- Standards based, open source enabled, build versus buy
- Empowers user communities, activism, large scale collaboration, shared infrastructure

Ways to think about truly data-centric opportunities

Think beyond end-to-end packaged software and SaaS solutions

- “Cognitive computing” platforms
- “Big data” platforms
- Repackaged legacy MDM
- Application-centric integration suites
- Data preparation suites
- Robotic or intelligent process automation
- Any other “AI-enabled” X, Y or Z that simply generates lot of market noise and gets in the way

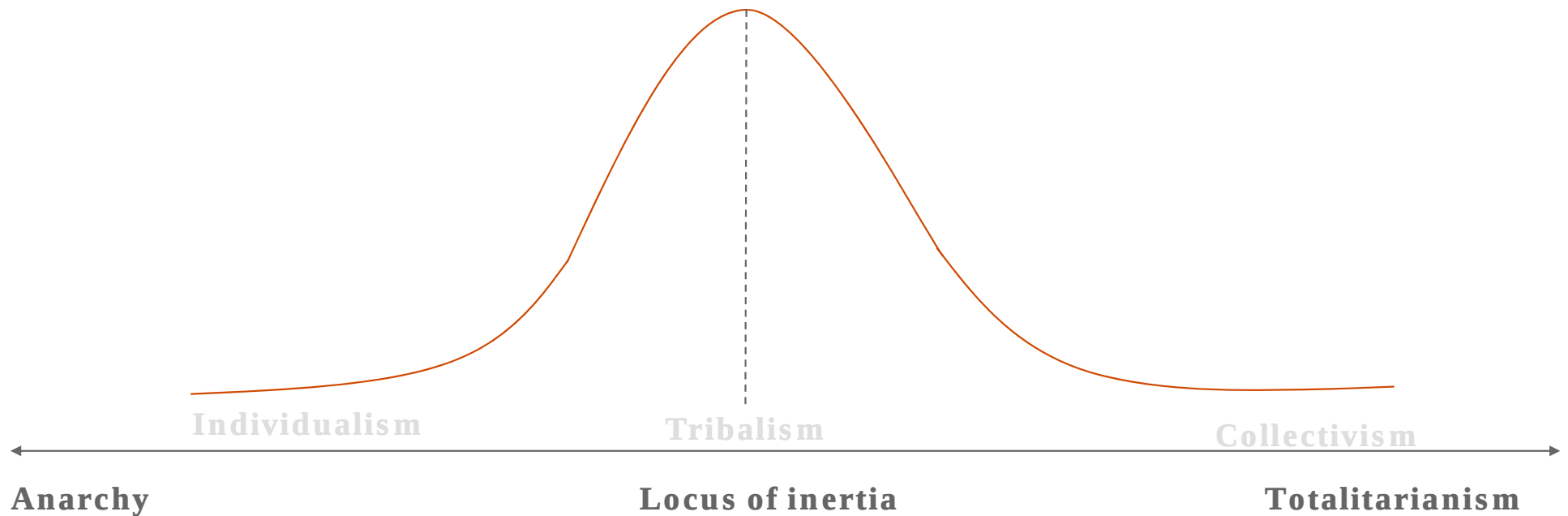
The elephant in the room is always organizational change. However, wrong technology and data strategy approaches prevent change also.

Those who succeed have effective strategies, means and execution all going for them *at the same time*.

Consider how to get the business and IT together building and solving foundational data problems

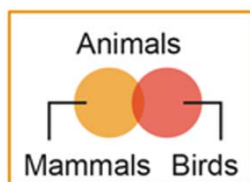
- Bespoke projects with worthy objectives and practical means of success
- Human-in-the-loop computing—constructing data-description feedback loops for knowledge foundations
- Software and SaaSes that do encourage better data modeling and relationship-rich integration
 - semantic graph databases
 - smart data hubs
 - NoSQL + SQL modeling—building bridges between tabular, document and graph
 - automated taxonomy and ontology generation, but as a starting point
 - Automated process mapping, but as a starting point
 - Clever modeling or visualization tools to encourage deeper, system-level understanding

The real inhibitors to adoption aren't technological – they're rooted in tribal biases and resistance to change



Tribalism – Machine learning edition

Symbolists



Use symbols, rules, and logic to represent knowledge and draw logical inference

Favored algorithm

Rules and decision trees

Bayesians

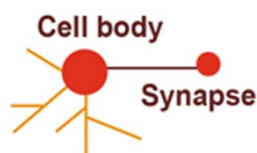


Assess the likelihood of occurrence for probabilistic inference

Favored algorithm

Naïve Bayes or Markov

Connectionists



Recognize and generalize patterns dynamically with matrices of probabilistic, weighted neurons

Favored algorithm

Neural network

Evolutionaries



Generate variations and then assess the fitness of each for a given purpose

Favored algorithm

Genetic programs

Analogizers



Optimize a function in light of constraints (“going as high as you can while staying on the road”)

Favored algorithm

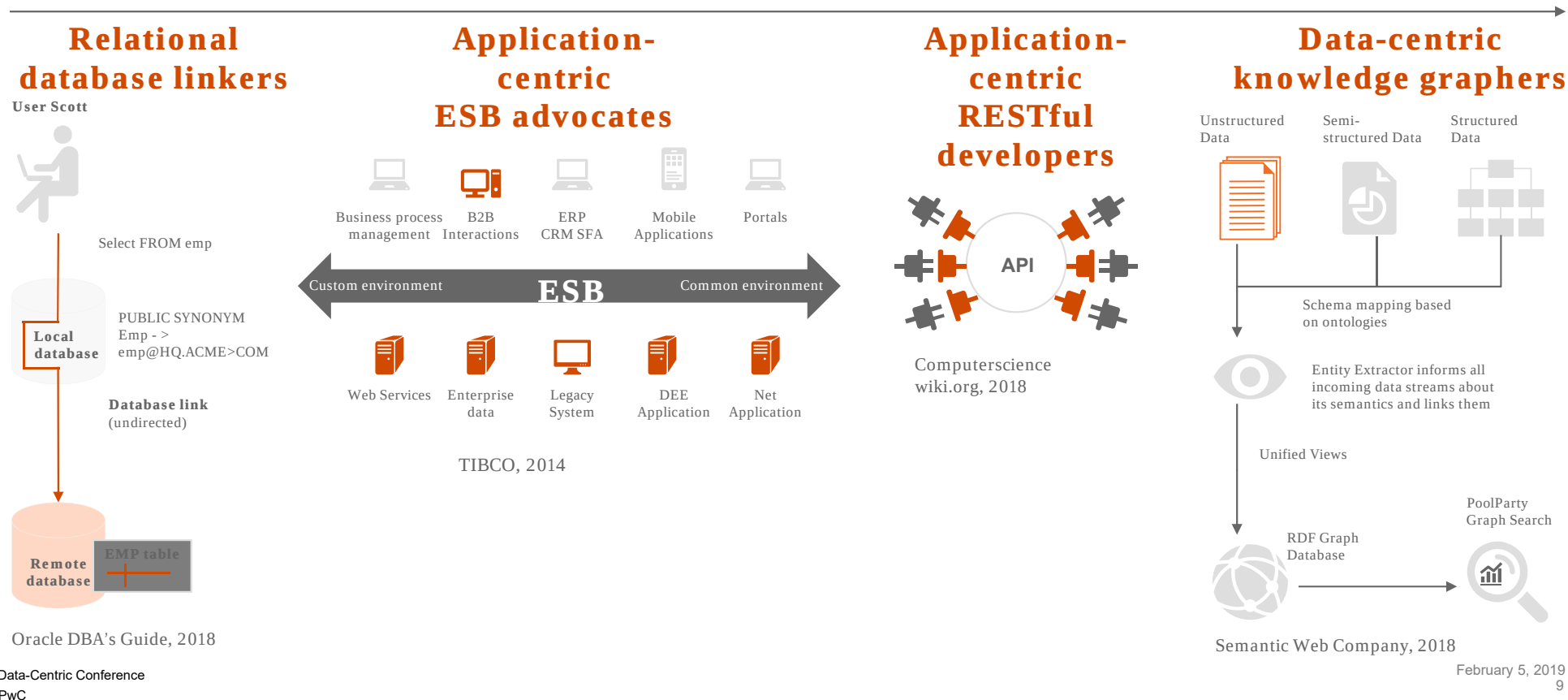
Support vectors

Source: Pedro Domingos, *The Master Algorithm*, 2015

More at “Machine learning evolution”: <http://usblogs.pwc.com/emerging-technology/machine-learning-evolution-infographic/>, PwC, 2017

Tribalism – Data integration edition

Trend toward more data centrality this way→



Crossing the chasm between the tribes

Reducing the amount of unfamiliarity developers confront--familiar document means to achieve comparable ends to graph:

- **Semantic suites that use the JSON format and familiar hierarchies:** SWC's PoolParty is an example
- **GraphQL:** A popular *document* shape language that talks to APIs using SELECT-like statements and tree shapes; retrieves only the data you need, provides needed feedback to API owners, helps with API sprawl
- **Accessible web as database methods:** JSON-LD and Schema.org, etc. vocabularies
- **Document “schemas” via data objects:** JavaScript objects to developers = documents to NoSQL DB types; Object Data Modeling instead of database semantics
- **Mongoose or MongoDB JSON schema features + GraphQL:** MongoDB object modeling and querying that can be used for subdocument filtering within a GraphQL context
- **HyperGraphQL:** A GraphQL UI for Linked Data, restricted to certain tree-shaped queries
- **Universal Schema Language:** Mike Bowers' document/graph query and modeling language still in development
- **COMN:** Concept and Object Modeling Notation, Ted Hills' well-defined NoSQL + SQL data modeling notation

Examples of data-centric approaches in 2019

Data cataloging and auditing

- Portal-style data asset visibility
- Data inventory and curation as a first step to privacy compliance
- Information supply chain mapping
- Back-end classification

Intelligent assistants

- Expanded user experience
- Cross-domain capability

Knowledge graphs

- Large-scale, heterogeneous integration for data discovery and asset tracking
- Platforms for advanced analytics and machine learning
- Knowledge bases for intelligent assistants

Smart data hubs

- Alternatives or adjuncts to data warehousing, or
- Integration across **both** operational and analytics data

Cybersecurity

- Threat intelligence, but active measures too
- Network analysis
- Identity verification

Off-chain to on-chain data quality for blockchain networks

- Ways to avoid garbage-in, garbage-out
- Supply chain integration
- Smart contracts for automated transactions and compliance
- Personal data protection
- **Self-sovereign identity**
 - Individuals have a **border** and **sovereignty** for personal data control in the same way countries exercise sovereignty within their borders
 - **Peer-to-peer relationship** status with other entities on the network

Sources: Kurt Cagle, "The Semantic Zoo," *Forbes*, 18 January 2019
Bridget Botelho, "What are the main features of data catalog software?" *TechTarget*, 10 May 2018
Christopher Allen, "The Path to Self-Sovereign Identity," *Life with Alacrity* blog, 25 April 2016
Phil Windley, "What is 'self-sovereignty'?" *Sovrin* Twitter video, 24 January 2019

2

Market buzz, trends
and dynamics to be
aware of

Emerging technologies and the data value chain

**IoT and
drone data
collection**

AI

AR/VR

**Blockchain
(immutable ledger sharing +
autonomous process coding)**

Data discovery

**Data
integration**

**Data
exploitation**

**Collect and
annotate**

Prepare

Organize

Integrate

Analyze

Visualize

**Make
decisions**

Create an inventory
of data sources and
the metadata that
describe them.

Enable access
to sources and set
up access-control
rules.

Identify syntax,
structure, and sem-
antics for each
data source.

Establish a common
data representation
of the data. Main-
tain data
provenance.

Analyze integrated
data.

Present analytic
results to a decision
maker as an inter-
active application
that supports
exploration and
refinement.

Determine what
actions (if any) to
take on the basis
of the interpreted
results.

Optimize

Plan

Manage
&
Monitor

Create

Execute

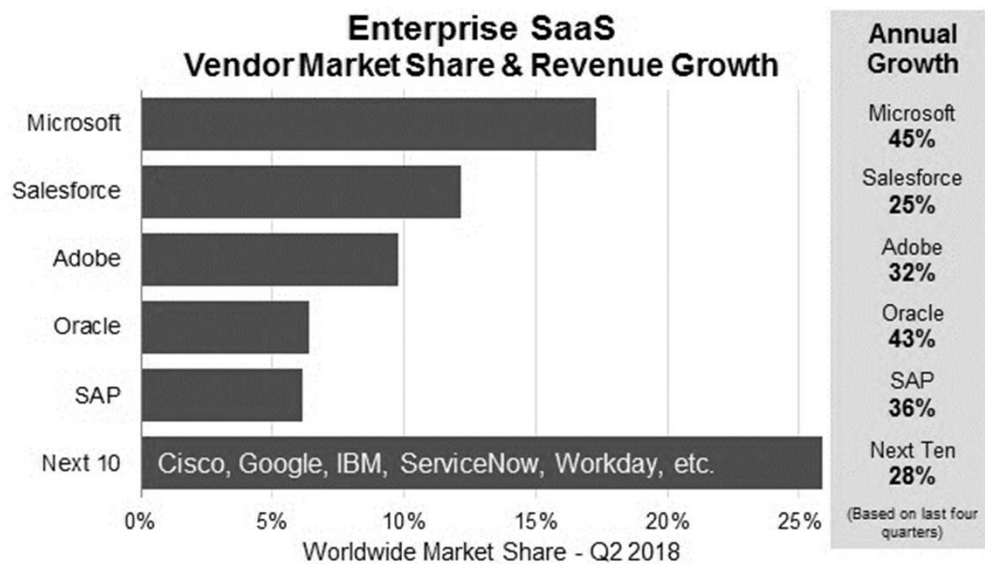
Refine

**Operational data
generation and
use**

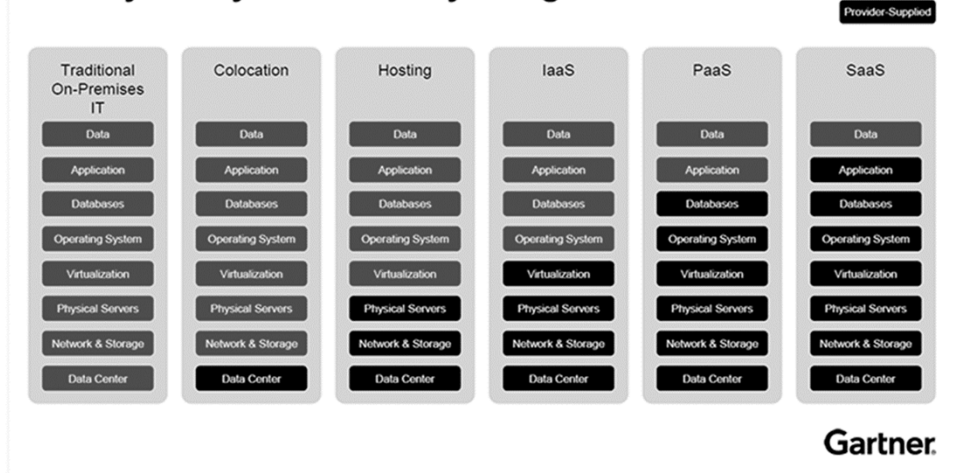
**3D printing output, IoT
distribution, robotics
and drone delivery**

PwC and <http://doi.ieeecomputersociety.org/cms/Computer.org/dl/mags/it/2013/01/figures/mit20130100571.gif>

SaaSes and clouds generally are incredibly popular. What are the implications?



Identify what you are already doing in the cloud



Trend toward owning and managing less and less of the stack

Enterprises used an average of 1,181 cloud services each by the end of 2017

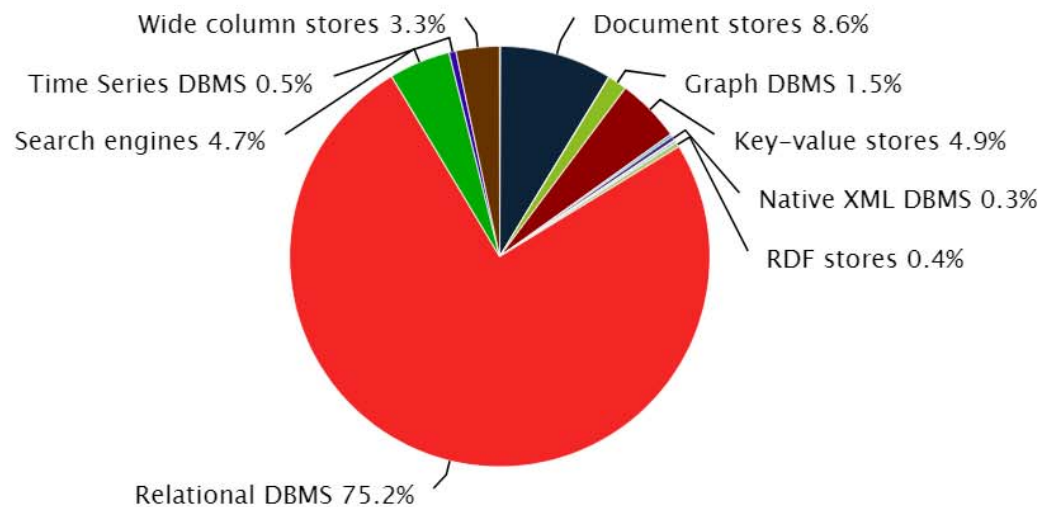
CATEGORY	# PER ENTERPRISE	NOT ENTERPRISE-READY
HR	139	95%
Marketing	121	97%
Collaboration	95	81%
Finance/Accounting	63	94%
CRM	62	94%
Social	24	89%
Cloud Storage	24	66%
IT Service/Application Management	22	97%

- Enterprises used nearly 1,200 cloud services each in Q4 2017, according to Netskope
- Most of these are SaaSes such as Salesforce, Workday, SAP Success Factors....
- Buy rather than build continues
- Even with enthusiasm for AI, data and analytics skills continue to be scarce

Netskope's *2017 Cloud Report*

Means of integration, and a database popularity ranking per DB-Engines

Ranking scores per category in percent, January 2019

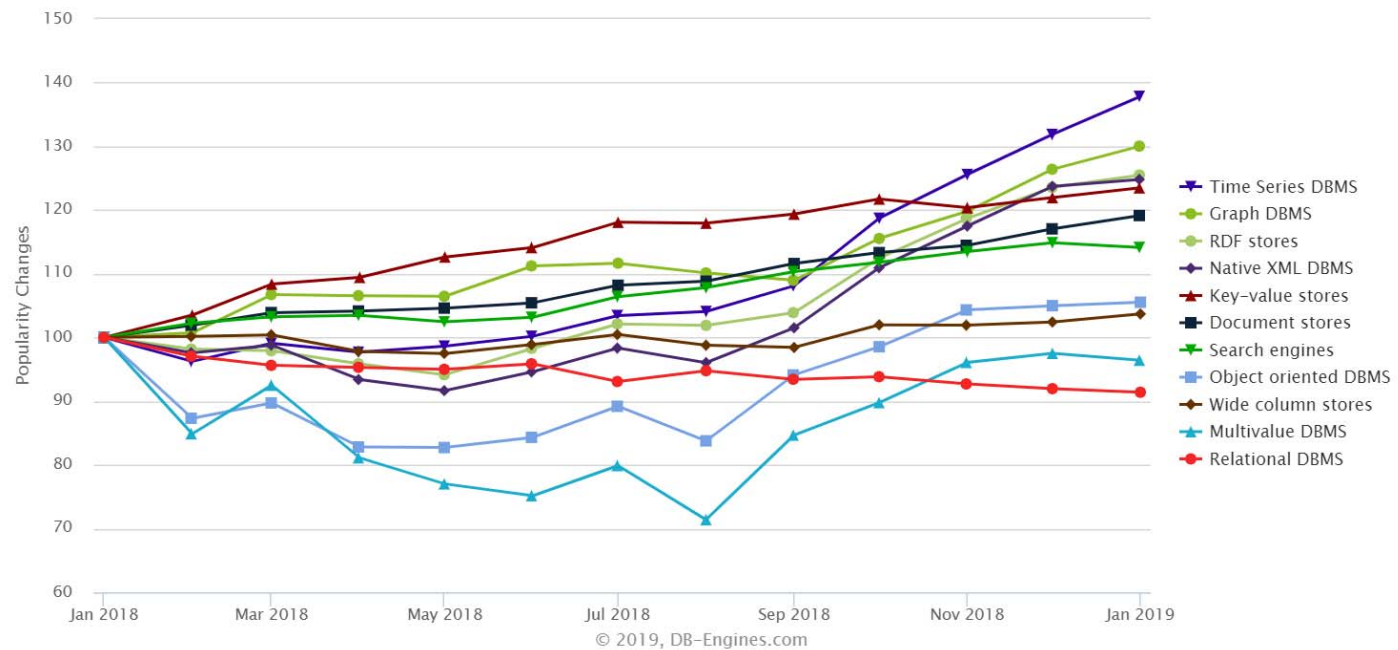


© 2019, DB-Engines.com

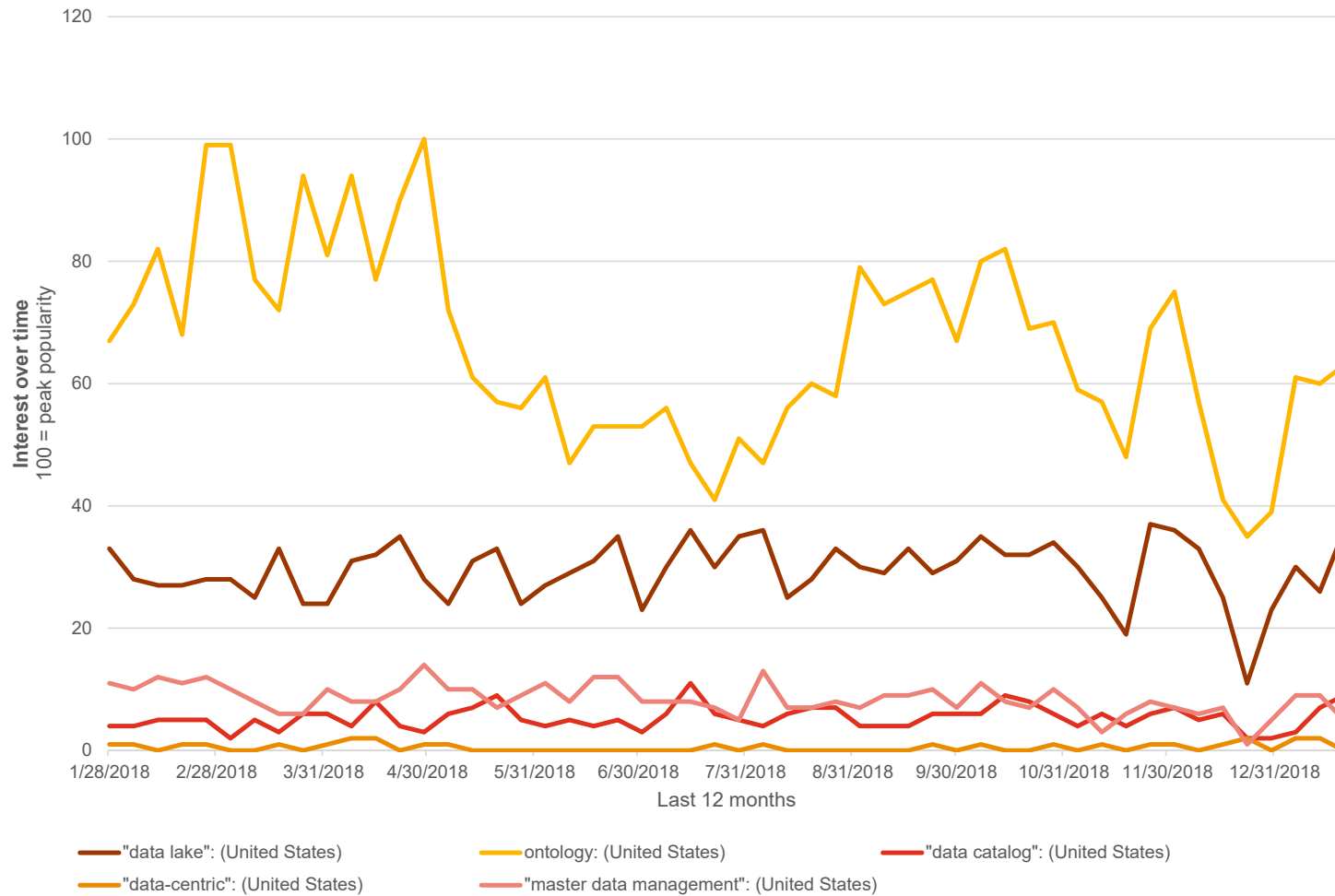
This chart shows the popularity of each category. It is calculated with the popularity (i.e. the [ranking scores](#)) of all individual systems per category. The sum of all ranking scores is 100%.

Database popularity by type (including model type)

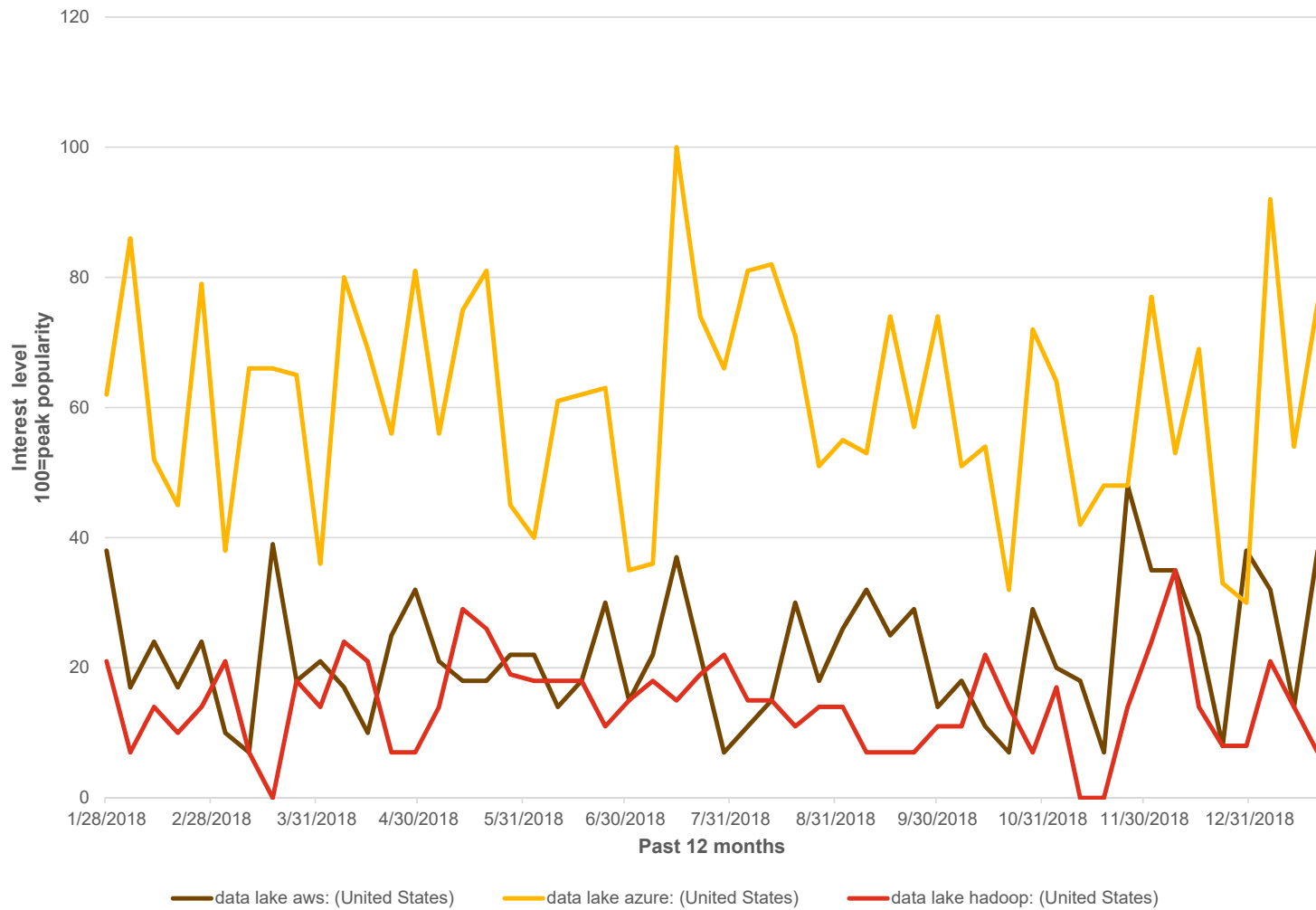
Trend of the last 12 months



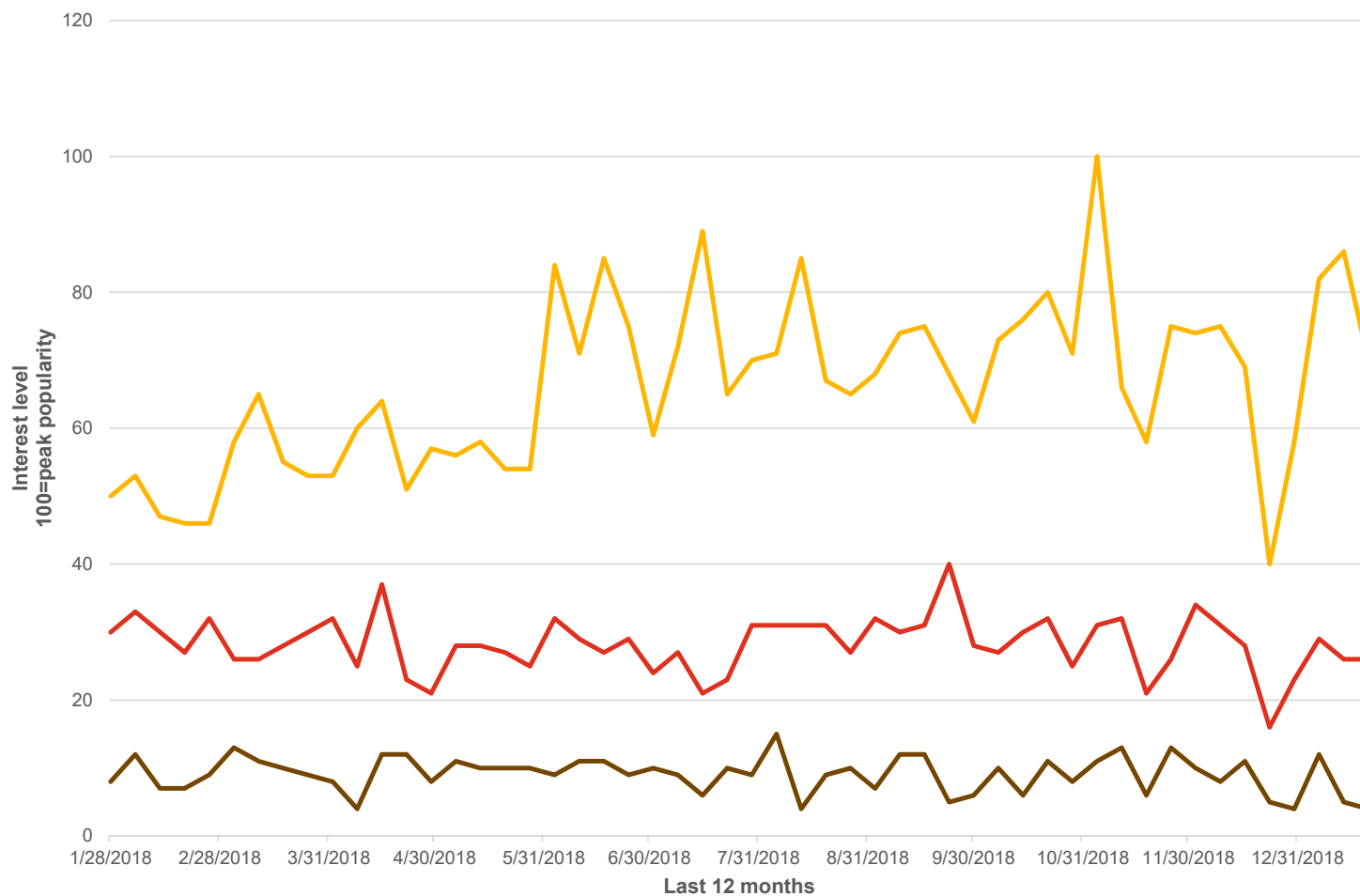
Popularity of the "O" word and "data lake" versus other data terms (per Google Trends)



Popularity of data lake + storage terms



Graph and related search term popularity



3

Current approaches
to the market

Data catalogs

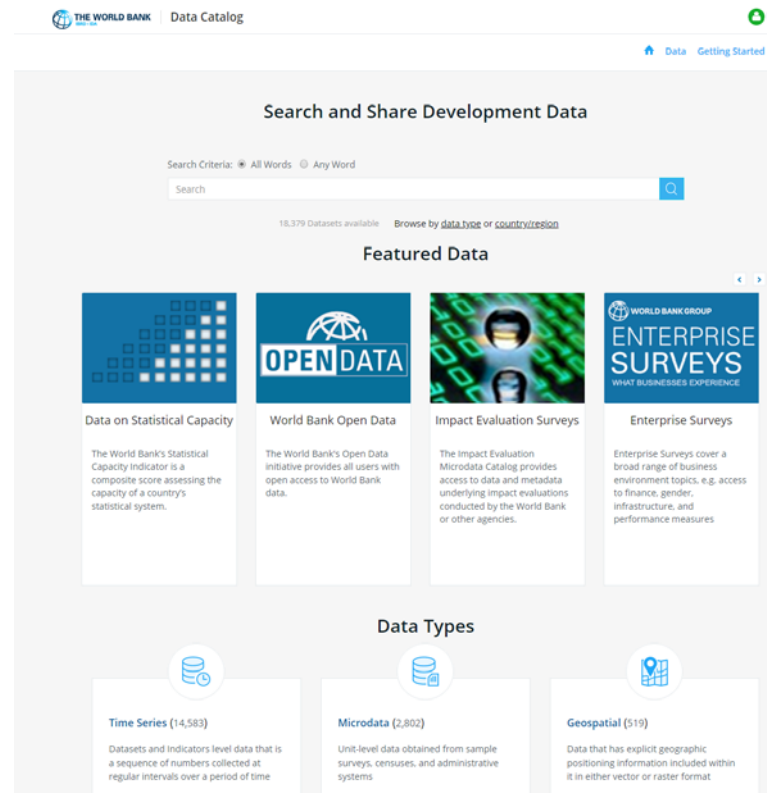
Open data initiatives provide examples to follow, but what about data audit?

Open data and data catalog development sites

- World Bank
- Data.gov
- Data.gov.uk
- OpenAfrica
- Data.world

Data-centric audit and protection (DCAP)

- Can be heavily vendor-driven (Protegrity and Informatica lead Gartner's ranking, e.g.)
- Bespoke methods would be more in line with data-centric build-vs.-buy principles



Cybersecurity: Will black box services give way to open graphs that ordinary analysts can use?

Gemini Data vs. Palantir

“Placed in the hands of an analyst, Gemini [Data] allows them to **start with an event**, such as an anomalous router message ^{ASM1} suspicious email address, and **then work out from there. The product’s GUI guides the analyst through possible connections to that particular piece of data**, allowing the analyst to quickly and iteratively explore how different pieces of data might be connected.”

--Alex Woodie, *Datanami*

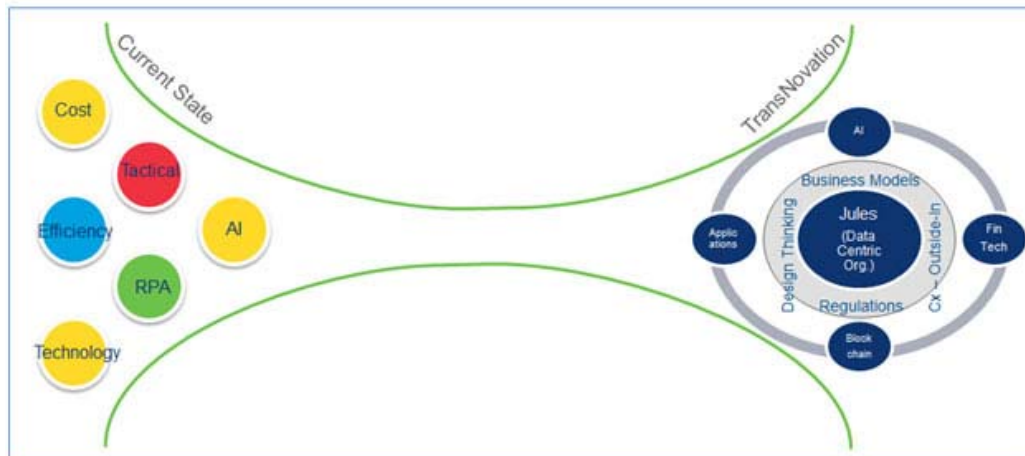
Cybersecurity at the DNC in 2016

“Take more vulnerable organizations that feel like they don’t have the resources. A good example from the book, the Democratic National Committee. OK? So before the election cycle gets going, they bring in Dick Clarke...he now runs a cybersecurity firm. **They do a quick survey** of the DNC’s computing system and they come back and **they basically say you guys are hopeless. OK? Like, you’re down in kindergarten levels....he showed them how much it was going to cost. And they said, great, this is too much money**, we’ll pay for it after the election. OK?”

And **then the FBI calls and says, by the way, the Russians are inside your system.** Well, I’m sorry. They called and they asked to be connected to somebody to who they could tell that to. **And they got connected to the help desk.**”

--David Sanger, author of *The Perfect Weapon*, as quoted on the *Council of Foreign Relations* website

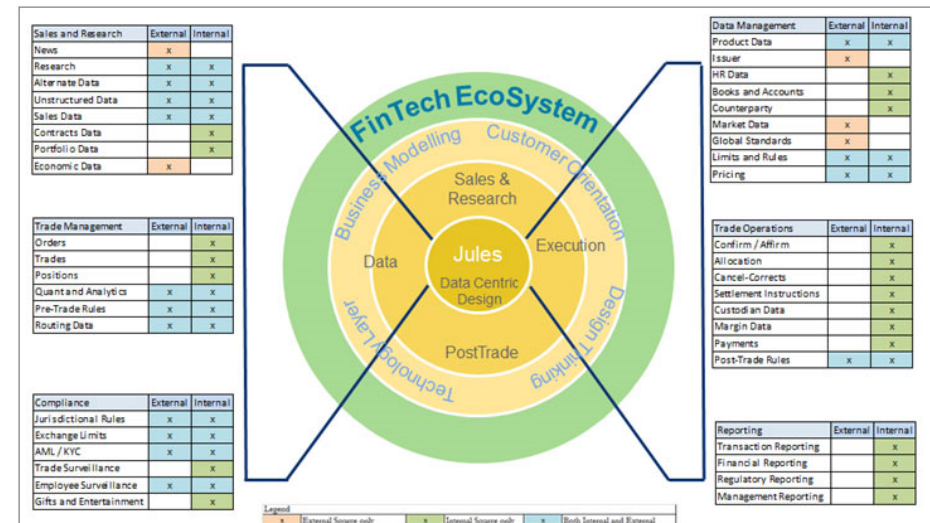
Data-centric architecture: Designing an investment bank from the inside out at Wipro



Highlighted features of Jules Data-Centric Design

- Applications pull data from Jules and push the processed data back to Jules
- Defined metadata model, lineage, ontologies, semantics
- Data controls, governance, stewarded centrally
- Data as a platform

Source: "Journey to the Centre of Data," Giridhar Vugrala, Managing Consultant – Capital Markets, Wipro, Sept. 13, 2018



Personal data protection:

A preliminary self-sovereign identity stack proposes JSON-LD as a messaging payload format

At the annual Internet Identity Workshop, members of the Decentralized Identity Foundation and the W3C Verifiable Claims Working Group cobbled together a standards-based self-sovereign identity (SSI) stack, which included JSON-LD, as well as two other options.

SSI is currently operational in the Sovrin Network, a public, permissioned blockchain run by 60 different stewards on six different continents. That network is largely standards based, particularly via the W3C and Decentralized Identity Foundation.

Sources: Oliver Terbu, "The Self-sovereign Identity Stack," *Medium* post, and the Sovrin website, 2019

Layer	Examples
Application	Selective disclosure, music app, rideshare service, extensions, etc.
Implementation	DIF Hubs, Indy Agents, uPort app, etc.
Payload	JSON-LD, JWT, CWT
Encoding	ProtoBuf, Cap'n Proto, MessagePack, JSON, CBOR, etc.
Encryption	Ciphersuites, JWE, etc.
DID AuthN	Key ownership, verification, challenge/response, etc.
Transport	QR Code, HTTP, BLE, NFC, FTP, SMTP, etc.
DID Resolution	DID -> DID Doc / service and key resolution
DID Operations	CRUD support for a DID Doc
Storage	Optional, separate storage of DID metadata, e.g., IPFS
Anchor	Bitcoin, Ethereum, Veres.One, Sovrin, etc.



[View original](#) [Flag media](#)

 **Joshua Shane** @elbowspeak 5d
Curious what comprises a "self-sovereign identity" solution stack?
Here's a great overview from @uport_me's Oliver Terbu
link.medium.com/AAftdYYQT

Dynamic knowledge graphs: Tracking virus mutations with the help of graph databases

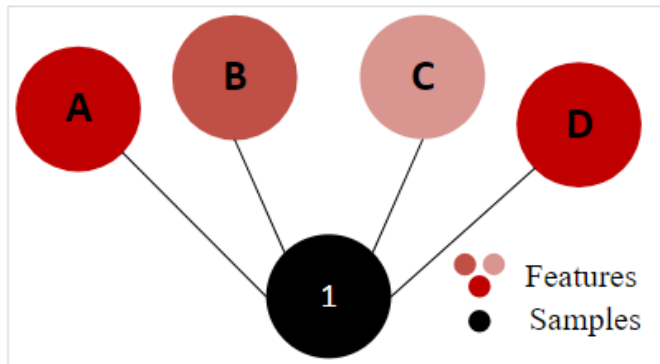


Figure 1 - Schematic representation of a sample with its feature in our graph database

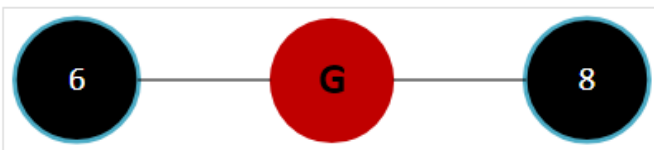


Figure 2- Two sample connected through a common feature

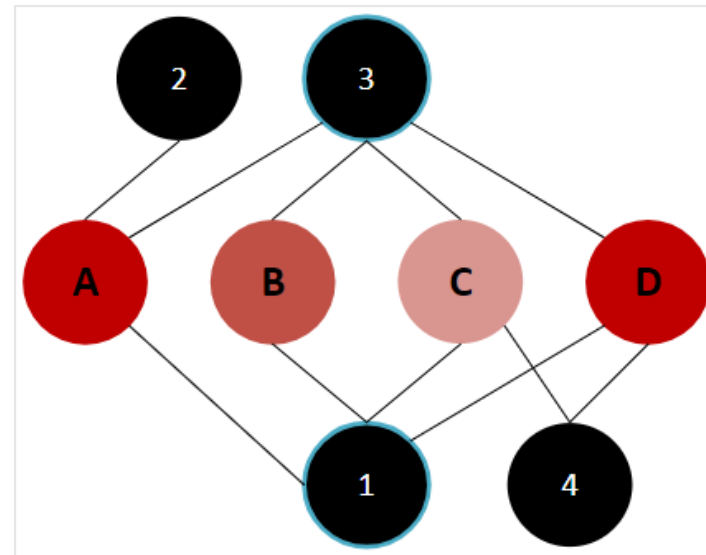


Figure 3 - Example: sample 1 and 3 have all features in common. Sample 4 is quite equal but not equal enough

Kadir Bölükbaşı, “One Graph to Find Them All,”
G Data Security Blog, 8 January 2019

Intelligent assistants: Amazon (Alexa) and Microsoft (Cortana) share resources, while Apple (Siri) contemplates its next move

Alexa + Cortana:

“With the new Alexa + Cortana world, one could reach across the limitations of each platform domains and access the power of each platform. This has a synergistic effect where **one could at a future date construct meta Skills/Apps that use features from both platforms simultaneously.... I see Voice First platforms as a uniform way to access AI-assisted ontologies, taxonomies, and domains. Apps and Skills can be seen as a domain but also an extended taxonomy.**”

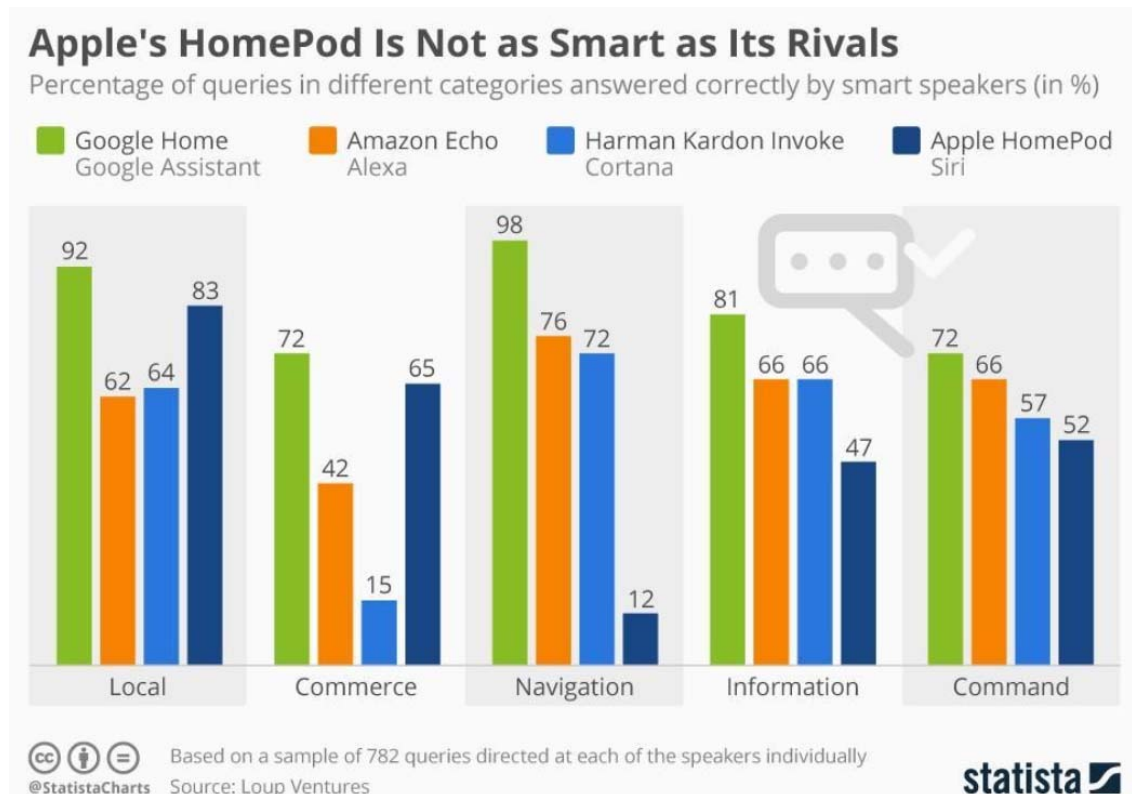
Source: Brian Roemmele, “Why Are Microsoft And Amazon Joining Forces With Cortana And Alexa?” Quora contributor on Forbes, Sep. 25, 2017

Siri:

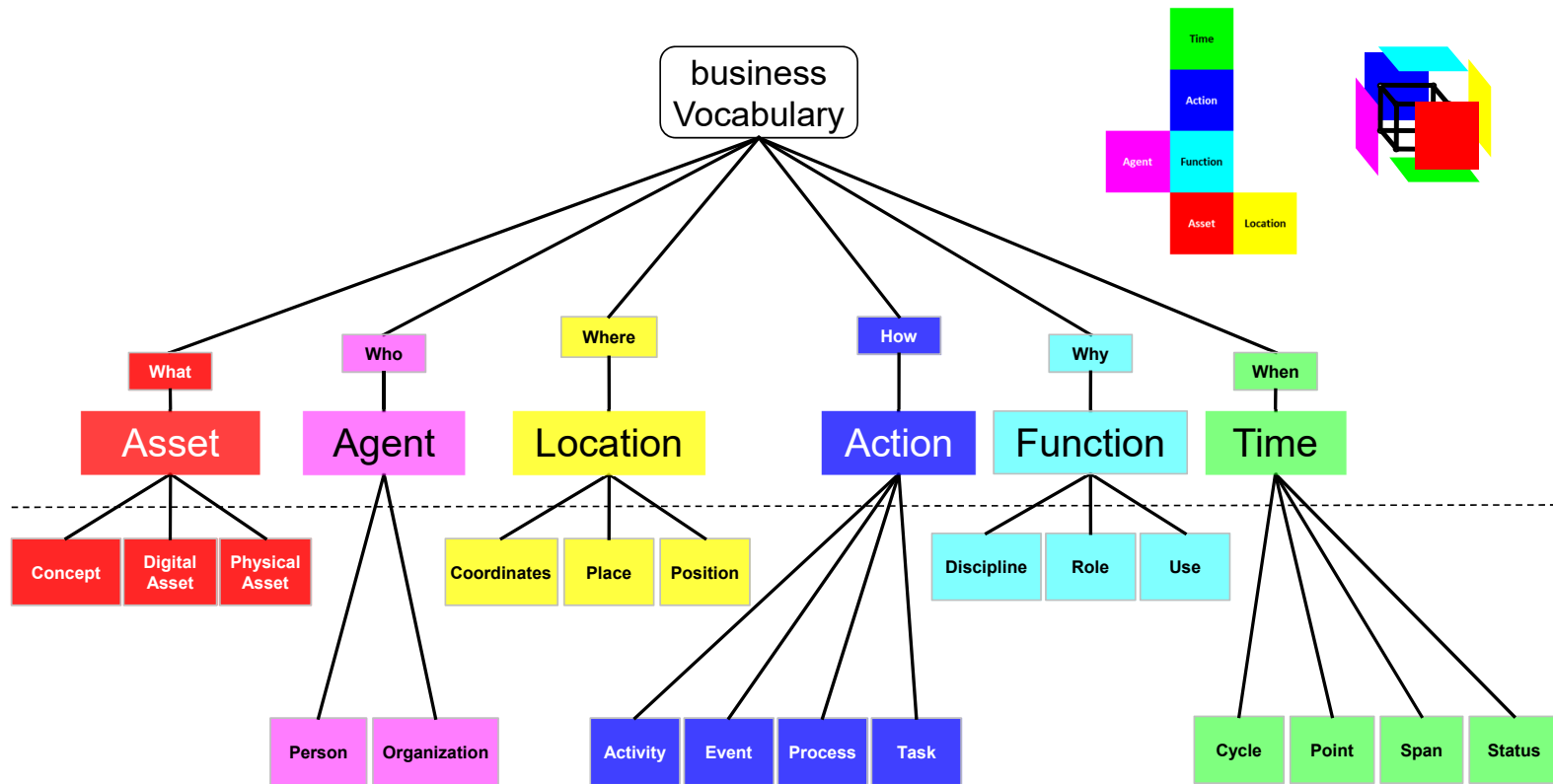
“**Apple executive Bill Stasior**, who has led the Siri team since joining the company in 2012, has been **removed as head of the project in a sweeping strategy shift favoring long-term research** over incremental updates, according to a report on Friday.”

Source: Mikey Campbell, “Apple removes Siri team lead as part of AI strategy shift,” *AppleInsider*, Friday, February 01, 2019, 02:47 pm PT (05:47 pm ET)

When will voice recognition accuracy reach 98 percent?
What happens when it does?
Should Apple re-hire Tom Gruber?



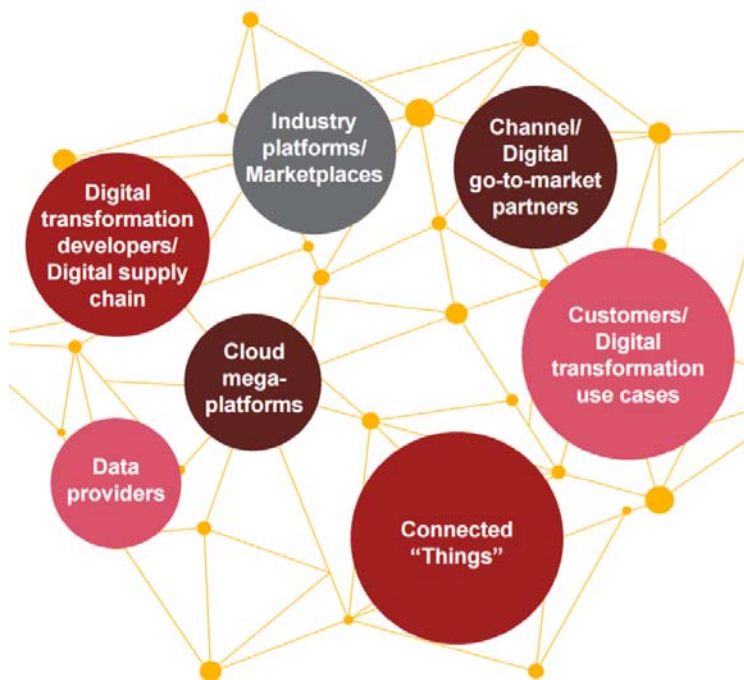
Data-centric simplified: QGFSA's Universal Information Classification



4

Case studies

With a knowledge graph base, companies can skate to new business models = Deep transformation



Source: Frank Gens, IDC Directions, 2017

- Once relationship data-enabled, organizations play different roles than they've been accustomed to in the digitized ecosystem.
- Some because of their data collection heritage can become data providers.
- Others take up roles in the data supply chain, or position themselves as industry platforms or marketplaces.
- Why are top companies able to cross industry boundaries?
- Why can unicorns extend the reach of their business models?

Largest changes in market cap by global company

Cross industry, 2018

		Company name	Location	Industry	Change in market cap 2009-2018 (\$bn)	Market cap 2018 (\$bn)
Known knowledge graph builders	1	Apple	United States	Technology	757	851
	2	Amazon.Com	United States	Consumer Services	670	701
	3	Alphabet	United States	Technology	609	719
	4	Microsoft Corp	United States	Technology	540	703
Operator of Taobao and AliBo: KG builder	5	Tencent Holdings	China	Technology	483	496
	6	Facebook	United States	Technology	383(1)	464
	7	Berkshire Hathaway	United States	Financial	358	492
Known KG builders	8	Alibaba	China	Consumer Services	302(1)	470
	9	JPMorgan Chase	United States	Financials	275	375
	10	Bank of America	United States	Financials	263	307

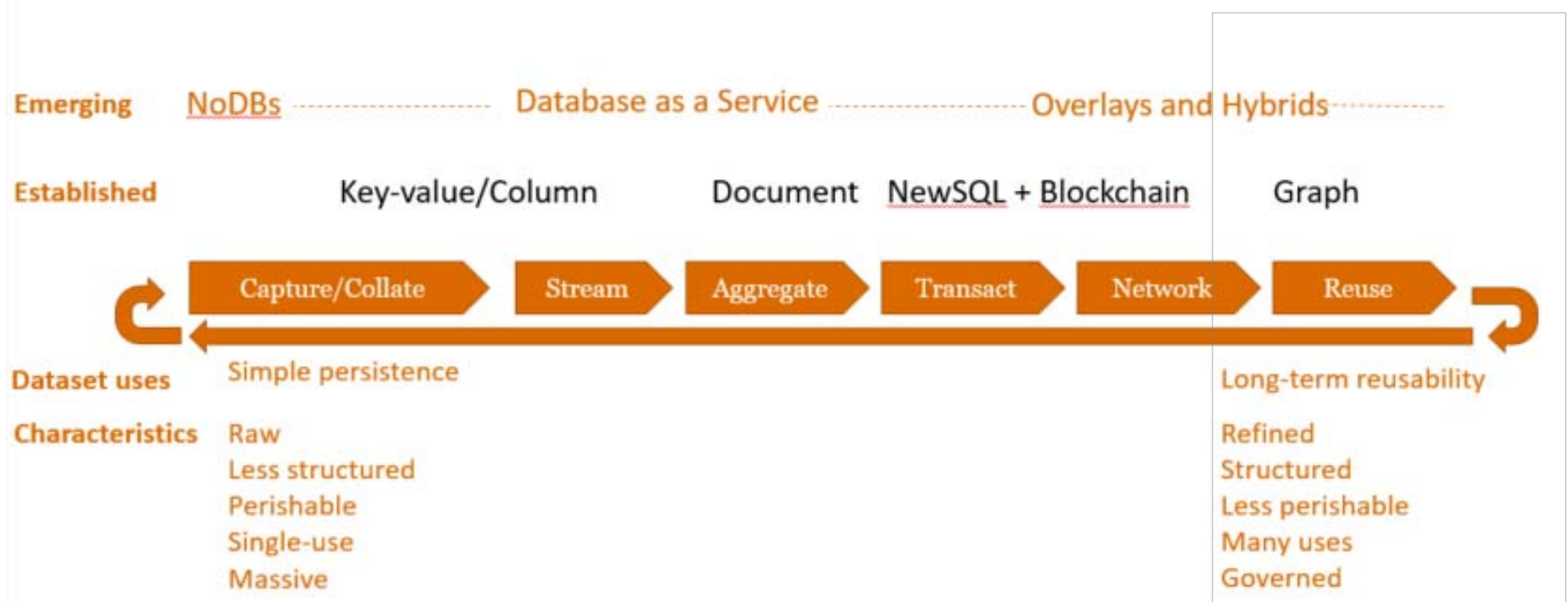
- IBM and Citi are also working on cross-enterprise knowledge graphs
- Many have cross-enterprise knowledge graph ambitions, but most are focused on a single use case
- S&P does cross-enterprise data management using relational tech

(1) Change in market cap from IPO date

(2) Market cap at IPO date

Source: Bloomberg and PwC analysis

Graphs (including hybrids) complete the picture of your transformed data lifecycle and how it's managed

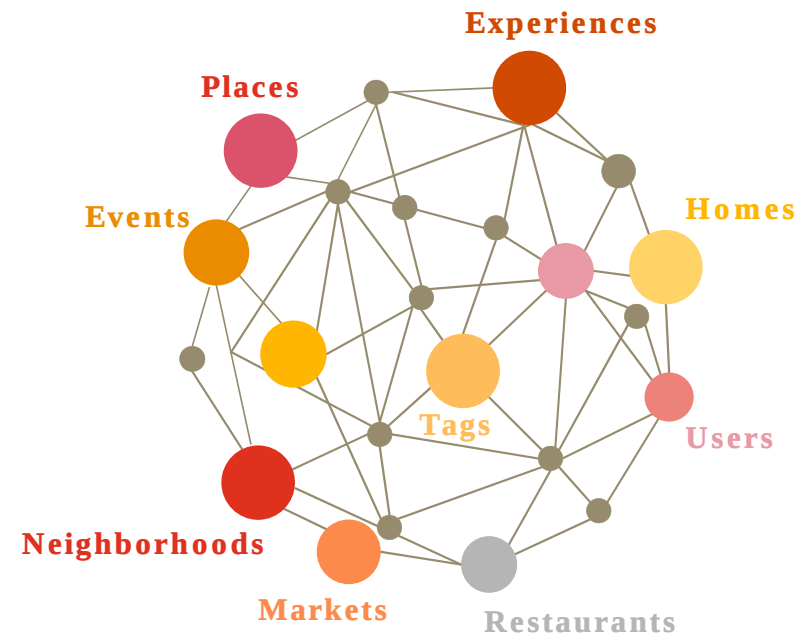


Transformation scalability – The AirBnB knowledge graph example

“In order to surface relevant context to people, we need to have some way of representing relationships between distinct but related entities (think cities, activities, cuisines, etc.) on Airbnb to easily access important and relevant information about them....

These types of information will become increasingly important as we move towards becoming an end-to-end travel platform as opposed to just a place for staying in homes. The knowledge graph is our solution to this need, giving us the technical scalability we need to power all of Airbnb’s verticals and the flexibility to define abstract relationships.”

--Spencer Chang, AirBnB Engineering



Airbnb Engineering, 2018

Most automated knowledge graph – Diffbot?

“Diffbot’s crawler regularly refreshes the DKG with new information and its machine learning algorithms are smart enough to pass over sites with histories of producing ‘logically inconsistent’ facts.

“‘That’s one of the reasons why we fuse information together from different sources,’ Tung said. ‘Our scale is such that there’s minimal potential for errors. We’d bet the business on it.’

“Diffbot launched in 2008 and counts 28 employees among its core staff of engineers and data scientists.”

--Mike Tung of Diffbot, quoted in *VentureBeat*

Diffbot claims an automated knowledge graph of 1 trillion + facts, designed to grow without humans in the loop.

That compares with 1.6 billion crowdsourced facts in Google’s knowledge graph, according to *VentureBeat*.

Kyle Wiggers, “Diffbot launches AI-powered knowledge graph of 1 trillion facts about people, places, and things,” *VentureBeat*, 30 August 2018

Versus more explicit, precise, contextualized meaning with a triadic, Peircean knowledge graph and less than 1M concepts?

“There are many different approaches for distinguishing a logical basis for ontologies, but Peirce basically says to base everything around 3s, explains [Mike Bergman of Cognonto]. That is,

- 1.the object itself;
- 2.what a particular agent perceives about the object;
- 3.and the way that agent needs to try to communicate what that is.

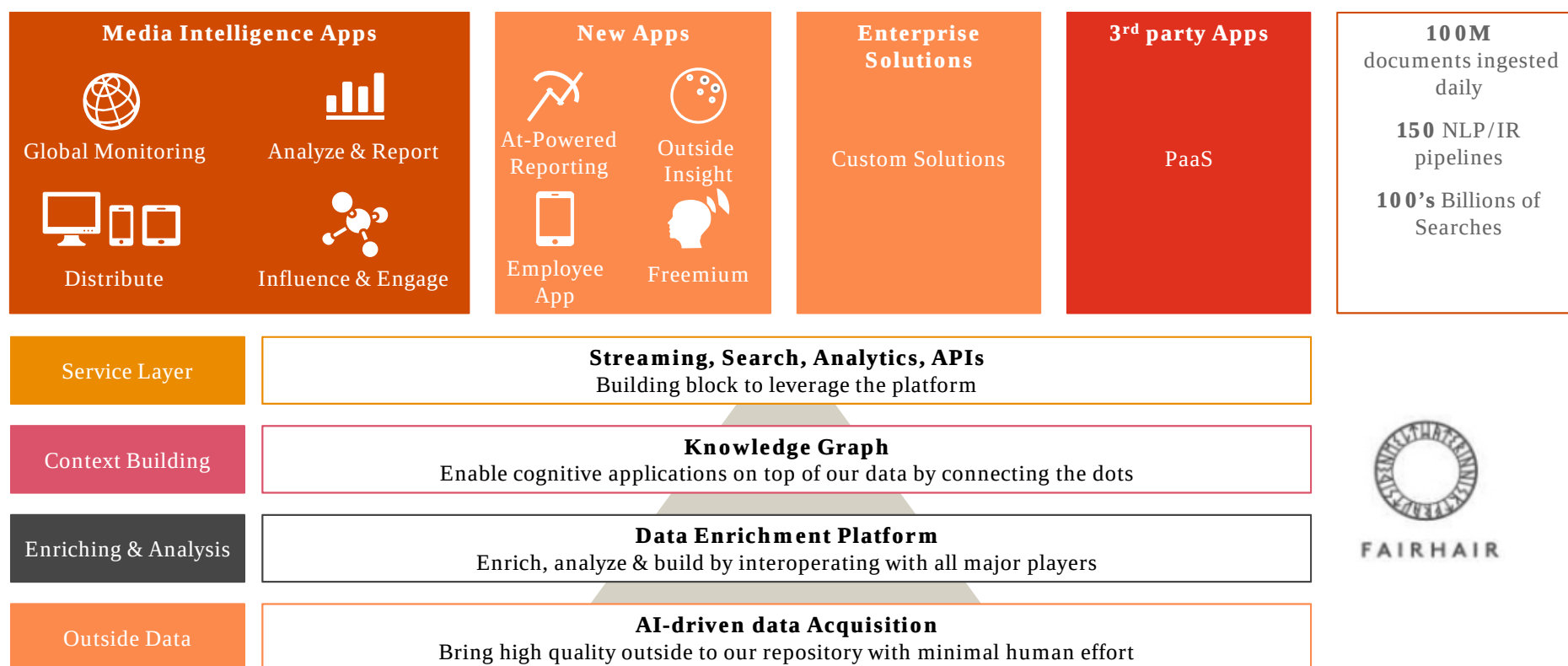
‘Without that triad it’s hard to ever get at differences of interpretation, context or meaning,’ he says, whether that be between something like events and activities or individuals and classes.

Once you adopt that mindset, a lot of things that seemingly were irreconcilable differences begin to fall away, and the categorization of information becomes really very easy and smooth....”

--Mike Bergman of Cognonto, quoted in *Dataversity*

Jennifer Zaino, “Cognonto Takes On Knowledge-Based Artificial Intelligence,” *Dataversity*, 23 November 2016

Contextual AI via a large knowledge graph at Fairhair.ai

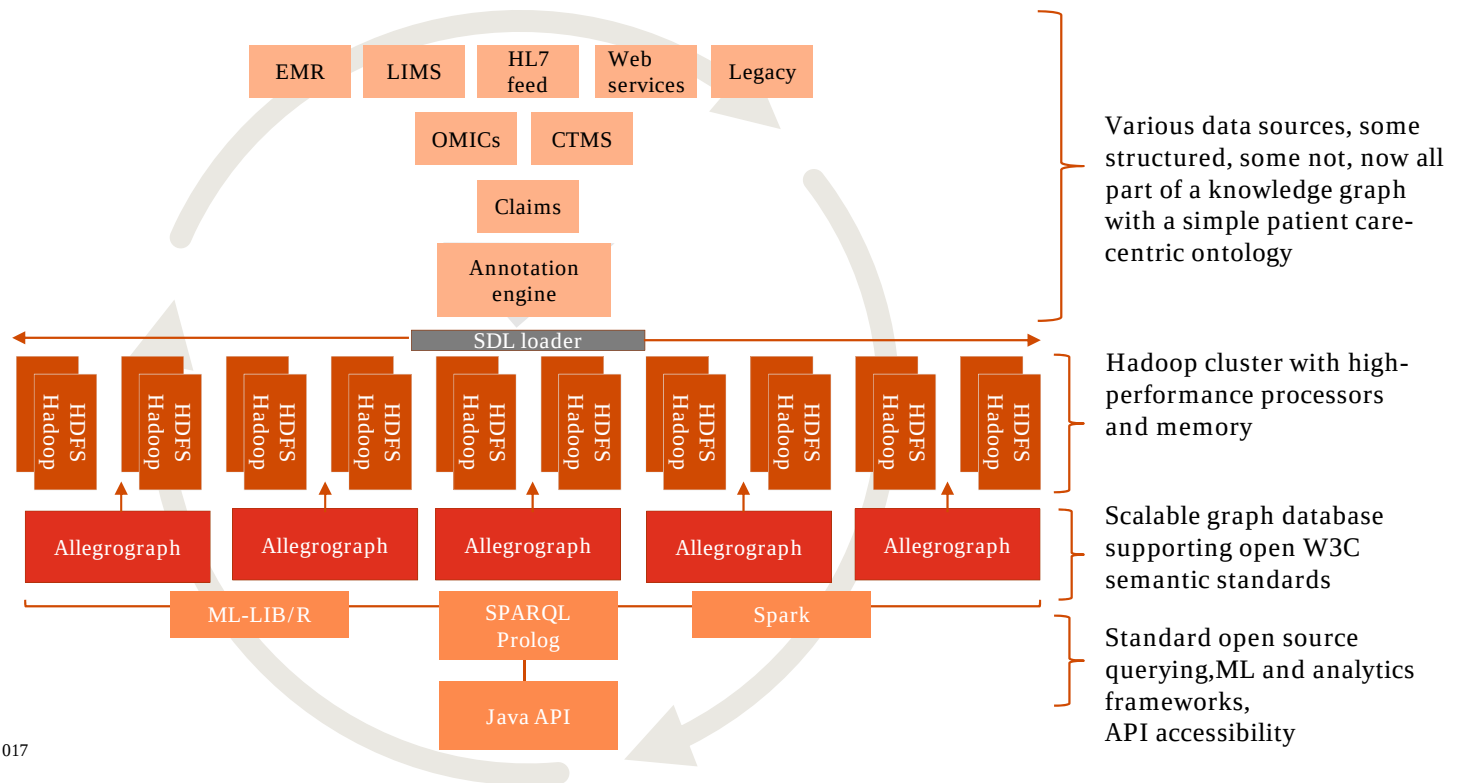


Montefiore's semantic data lake

Doctors can query the graph or harness ML + analytics and receive answers from the system at the point of care via their handhelds.

The system also acts as a giant feedback-response or learning loop which learns from the data collected via user/system interactions.

Montefiore Health, Franz, Intel and PwC research, 2017

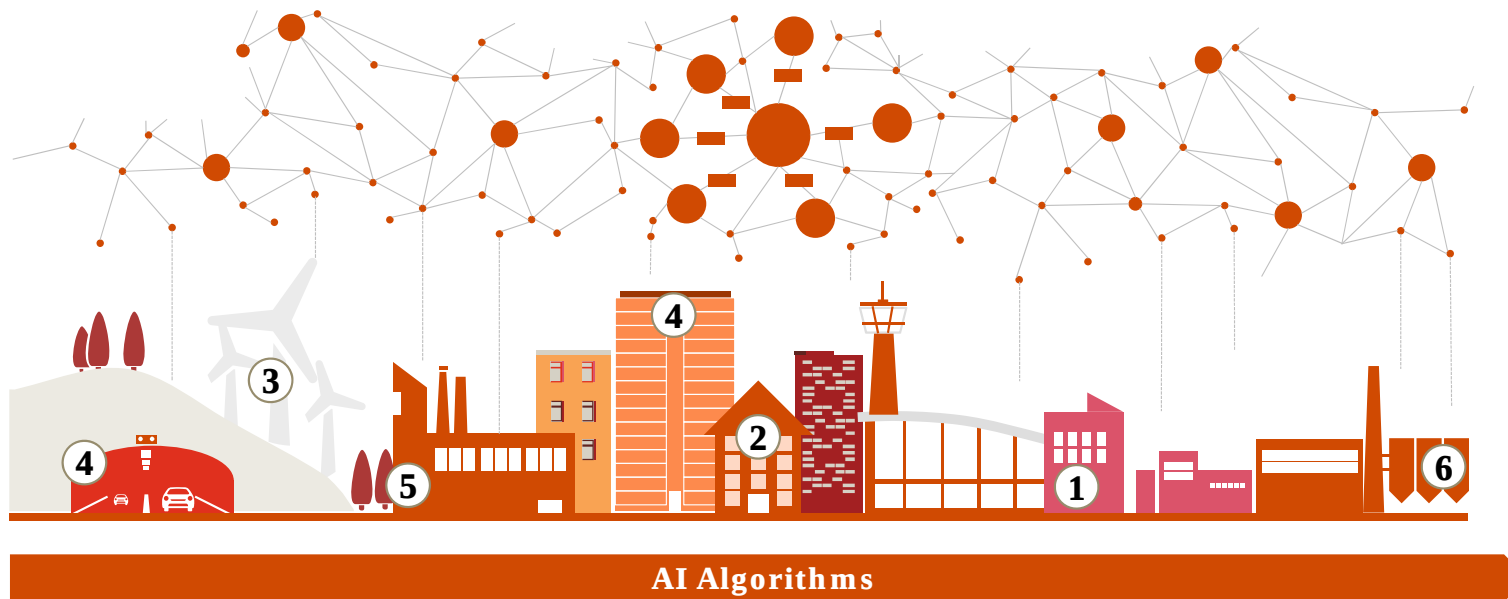


Siemens' industrial knowledge graph

Industrial Knowledge Graph

“Deep learning fails when it comes to context. Knowledge graphs can handle context and enable us to address things that deep learning cannot address on its own.”

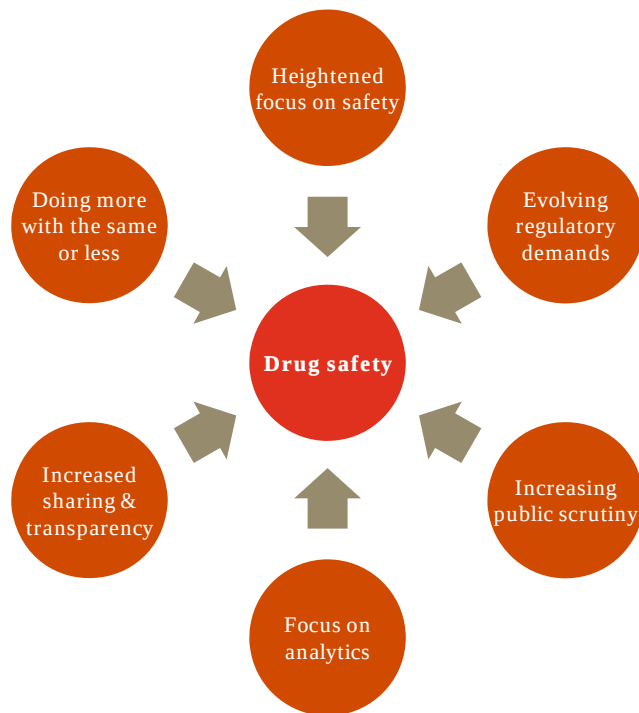
--Michael May, Head of Company Core Technology, Data Analysis and AI, Siemens



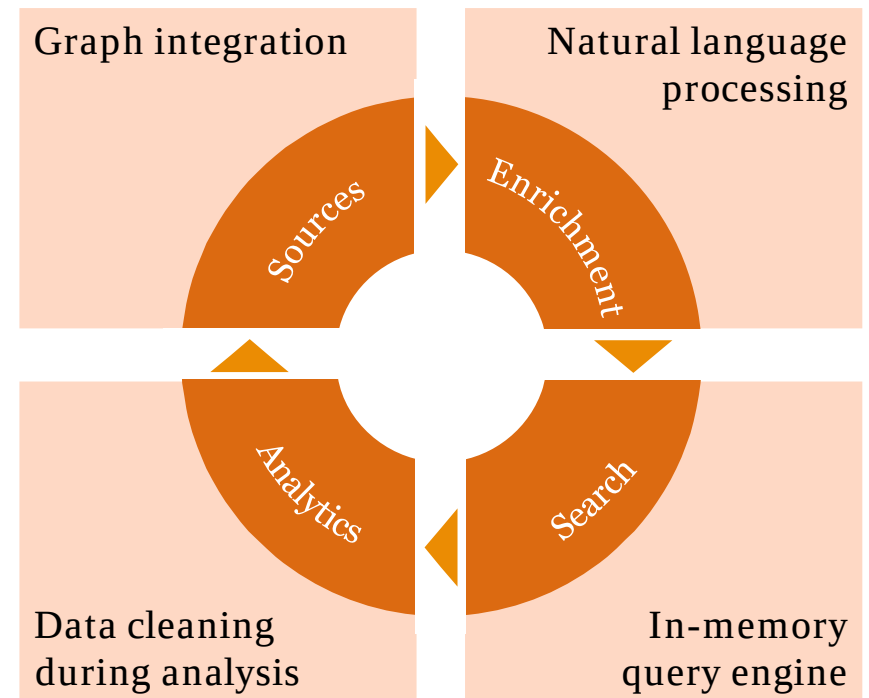
- 1 **09:00 – Analyze**
Turbine data hub
- 2 **11:00 – Configure**
Configure turbine
- 3 **12:00 – Maintain**
Master data Mgmt.
- 4 **13:00 – Mitigate**
Financial Risk Analysis
- 5 **15:00 – Contact**
Expert & Communities
- 6 **18:00 – Guide**
Rules & Regulations

Pharma knowledge graphs for patient safety

Challenges



Solutions



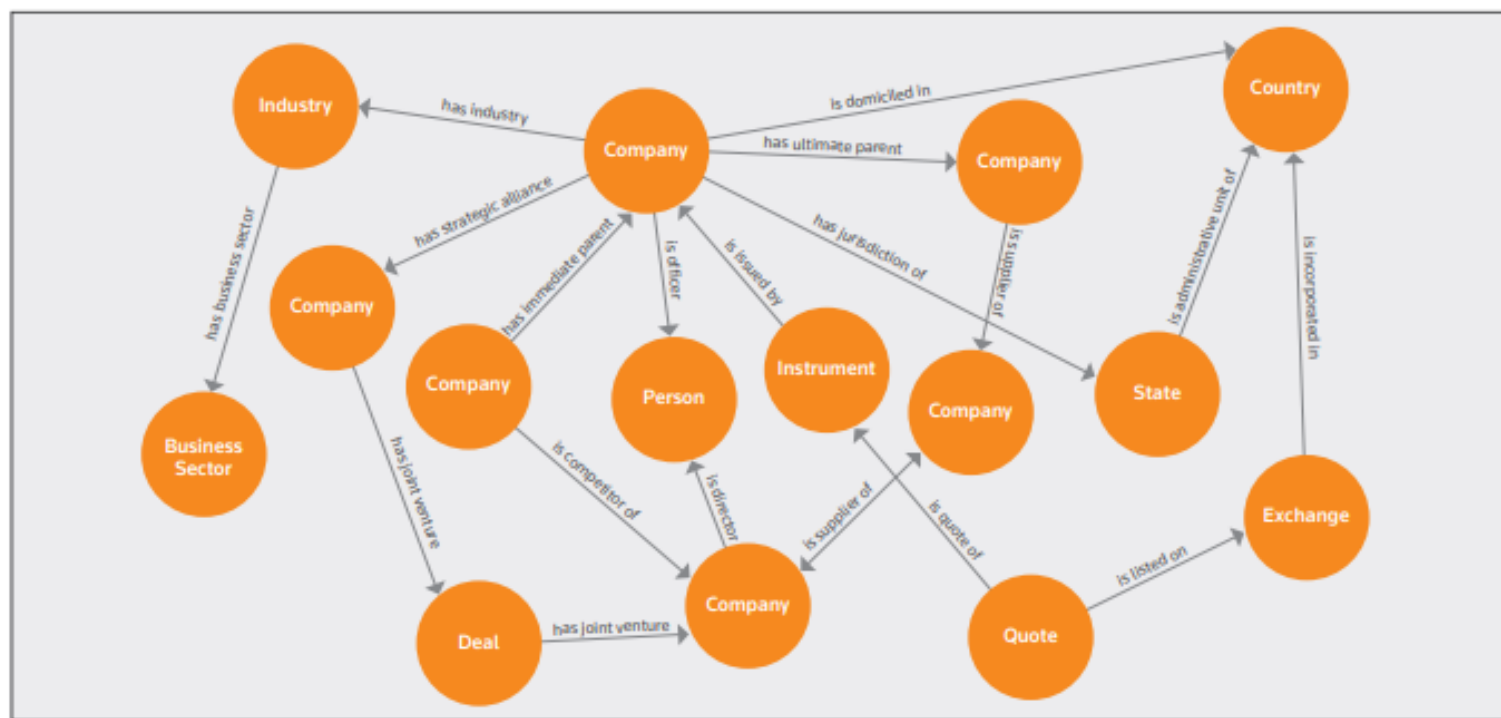
PwC and Cambridge Semantics, 2018

NuMedii's precision therapeutics knowledge graph



Ontotext and NuMedii, 2018

Thomson Reuters' financial knowledge graph as a service



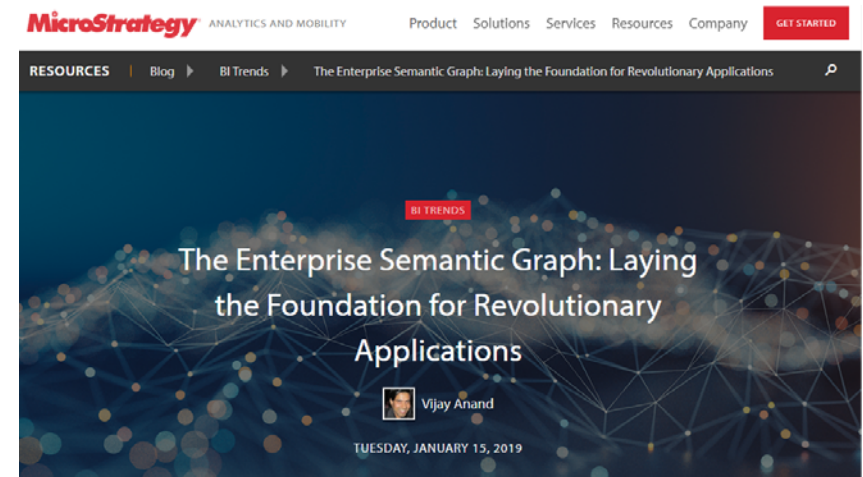
Thomson Reuters, 2018

Data-centric, or product-centric?

“MicroStrategy 2019 introduces the industry’s first Enterprise Semantic Graph.

- It elevates the potential of enterprise data assets, makes true federated analytics possible, and delivers personalized insights based on who you are, where you are, and what you’re doing.
- It delivers powerful search capabilities on top of all business information systems or data assets, making it incredibly easy to find insights.
- **It categorizes and federates each of your data investments in real time, constantly enriching the index with location intelligence and usage telemetry.**
- It delivers the underlying strength to fuel AI experiences for every role—with smart recommendations on authoring actions for analysts who build dashboards, to smart suggestions on content for business users who are looking for new insights.”

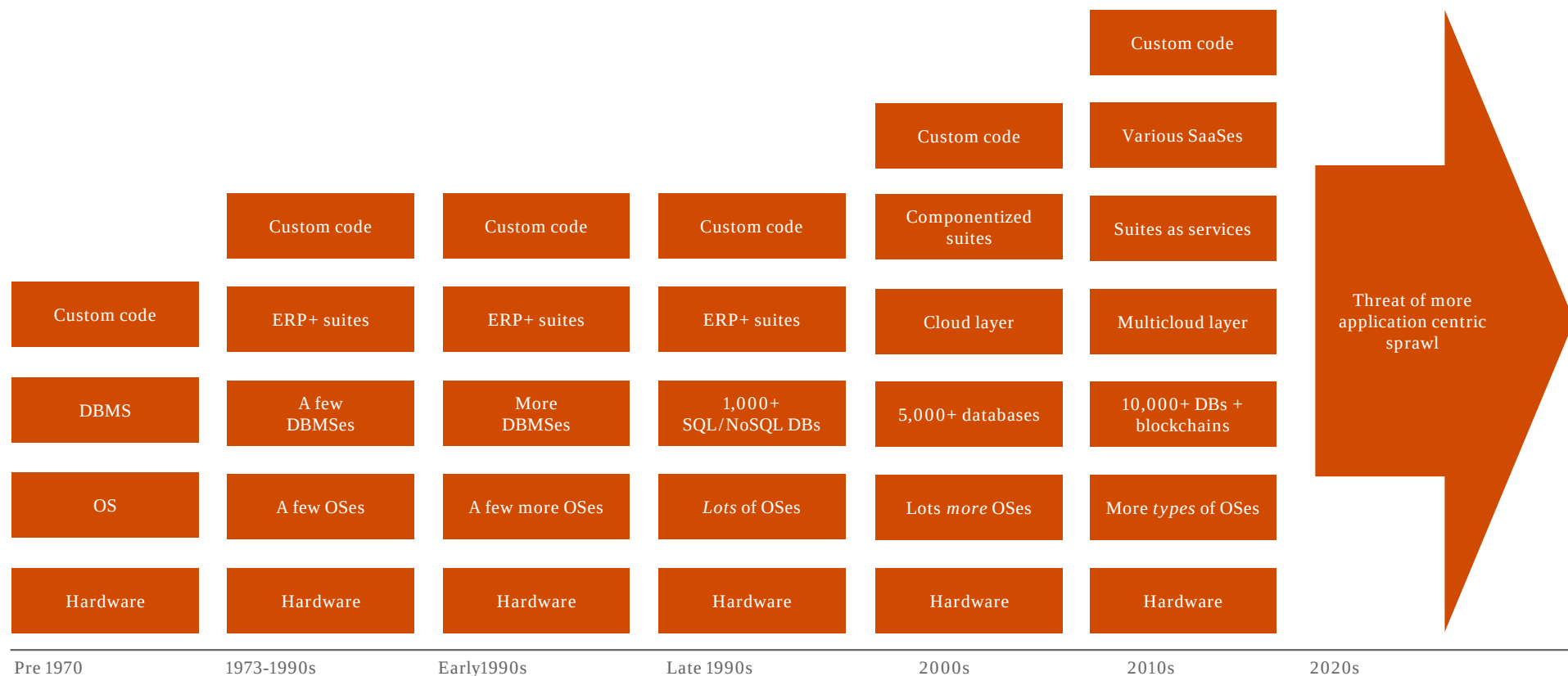
--Vijay Anand,
Microstrategy blog, January 15, 2019



5

Conclusion

The problem: System-level complexity and disconnectedness (product- and app-centric sprawl)



The key opportunity: Large-scale integration and model-driven AI

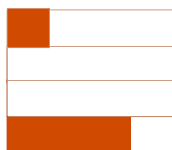
Previously dominant

Rule-based systems (includes KR)

“Handcrafted knowledge” is the term DARPA uses; rule-based programming + procedure replication in process automation, + some knowledge representation (KR)

- Strong on logical reasoning in specific concrete contexts
 - Procedural + declarative programming + set theory, etc.
 - Deterministic
- Can't learn or abstract
- Still exceptionally common and useful

Perceiving
Learning
Abstracting
Reasoning



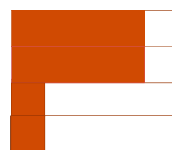
Example: Consumer tax software

On the rise and rapidly improving

Statistical machine learning

- Probabilistic
- From Bayesian algorithms to neural nets (yes, deep learning also)
- Strong on perceiving and learning (classifying, predicting)
- Weak on abstracting and reasoning
- Quite powerful in the aggregate but individually (instance by instance) unreliable
- Can require lots of data

Perceiving
Learning
Abstracting
Reasoning



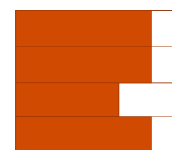
Example: Facial recognition using deep learning/neural nets

Nascent, just beginning

Contextualized, model-driven approach

- Contextualized modeling approach—allows efficiency, precision and certainty
- Combines power of deterministic, probabilistic and description logic
- Allows explanations to be added to decisions
- Accelerates the training process with the help of specific, contextual human input
- Takes less data

Perceiving
Learning
Abstracting
Reasoning



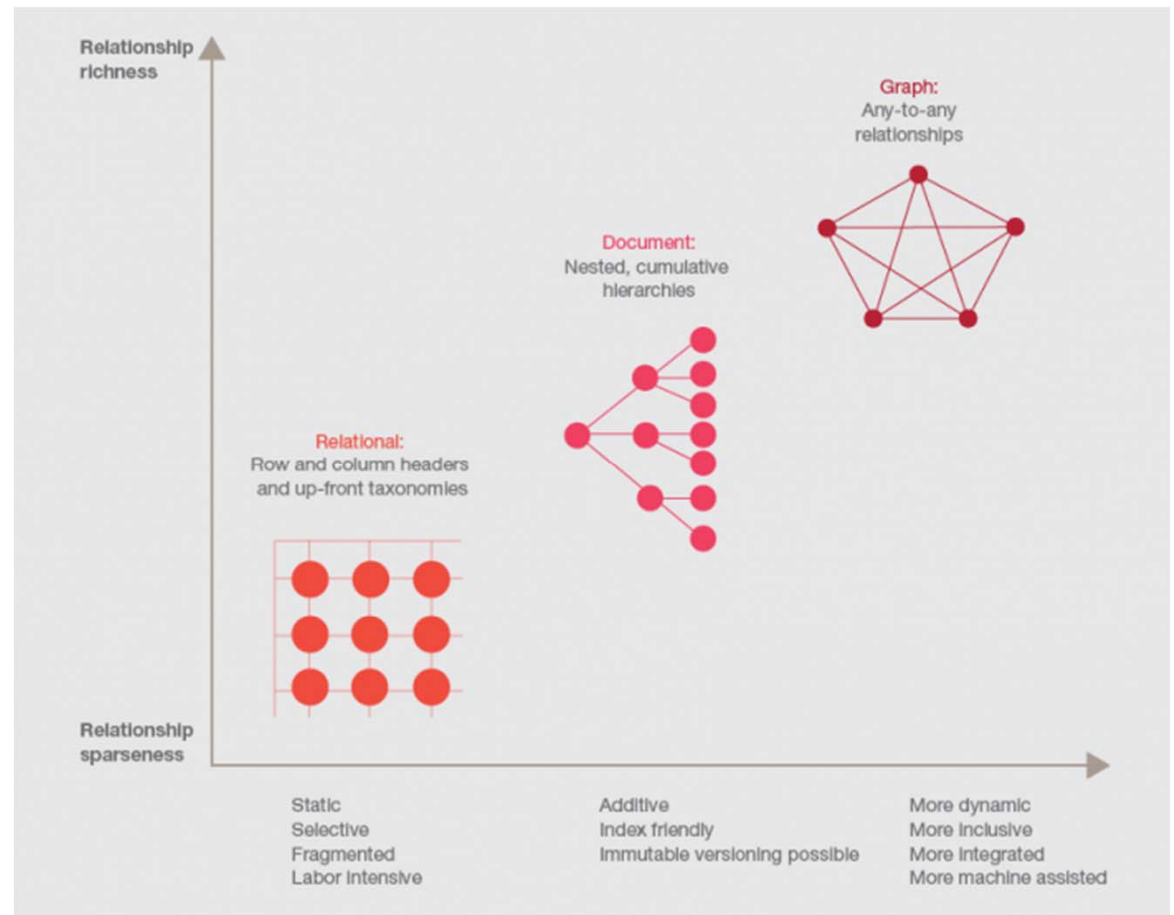
Example: Explains first how handwritten letters are formed so machines can decide—less data needed, more transparency.

John Launchbury of DARPA (<https://www.youtube.com/watch?v=N2L8AqkEDLs>), Estes Park Group and PwC research, 2017

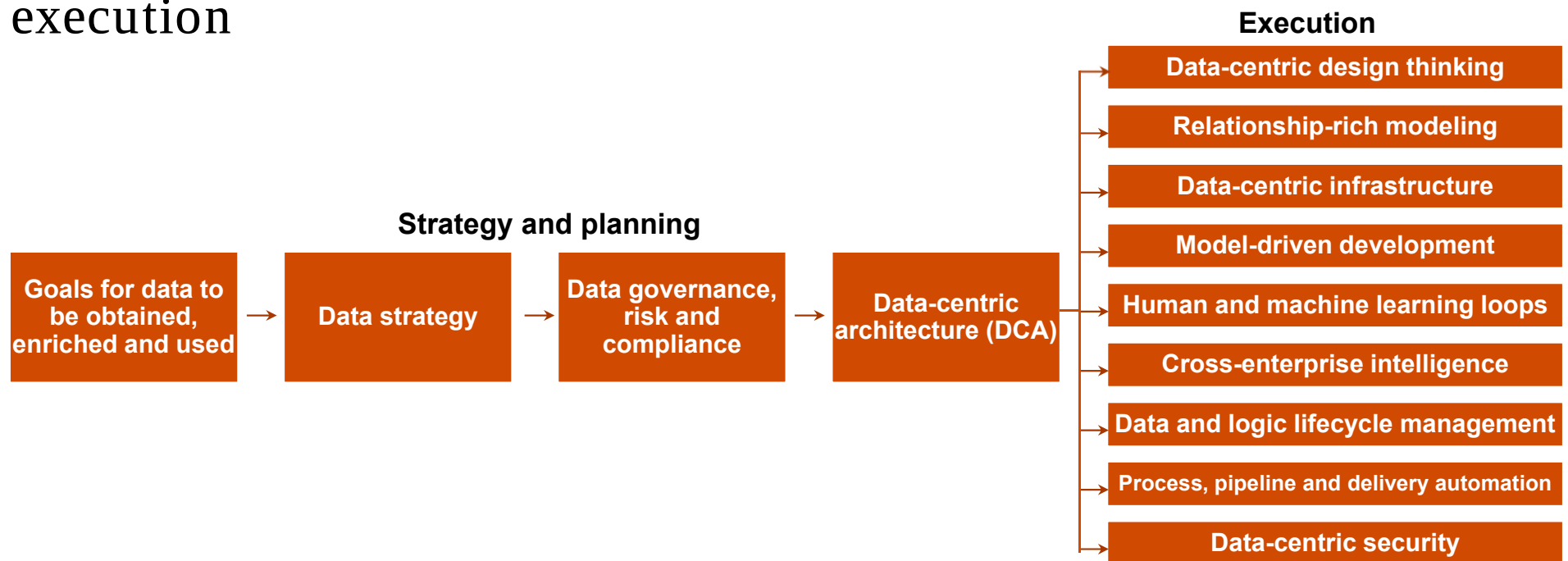
The key means: The right level of relationship richness

Use tables, document trees and graphs.

- **Graphs** articulate **relationship-rich** data
- **Tables:** Relationships are what's missing from most large-scale data, but table are too useful and human-friendly to ignore at smaller scale
- **Document trees** (e.g., taxonomies) are a **stepping stone** to graph models
- **Graphs are the parents** that bring the data model family all together, Tinker Toy-style
- **Bottom line:** A **machine readable, extensible model** of your organization
- **Build and maintain your advanced analytics/AI foundation** with that graph model



How to get started: Data-centric strategy, planning, architecture and execution



- In the data-centric view, every IT category is subordinated to centrally managed, model-driven data via data strategy, GRC and data-centric architecture (DCA)
- Relationship-rich modeling leads development for reasons of efficiency and effectiveness
- Standards based, open source enabled, build versus buy
- Empowers user communities, activism, large scale collaboration, shared infrastructure

Questions or comments?

pwc.com

Alan Morrison

Sr. Research Fellow

PwC | Integrated Content | Emerging Tech

Mobile: +1(408) 205 5109

Email: alan.s.morrison@pwc.com

PricewaterhouseCoopers LLP

488 Almaden Blvd., Suite 1800

San Jose, CA 95110 USA

<https://www.linkedin.com/in/alanmorrison>

@AlanMorrison [quora.com/profile/Alan-Morrison](https://www.quora.com/profile/Alan-Morrison)